

Lecture Notes on Differential Equations and Transform Techniques

Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.
8. Alan V. Oppenheim, Alan S. Willsky, with S. Hamid Nawab, *Signals and Systems*, 3rd edition, Pearson, 2023.
9. Simon Haykin, Barry Van Veen, *Signals and Systems*, 2nd edition, Wiley, 2003.
10. Steven H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*, 2nd edition, Westview Press, 2018.
11. Norman S. Nise, *Control Systems Engineering*, 7th edition, Wiley, 2015.

EE1060: Differential Equations and Transform Techniques**Lecture Notes - 36****Topics covered :**

1. Homogeneous system of Differential Equations
2. Wronskian and Fundamental set of Solutions

- Consider a system of first-order differential equations

$$\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu} \text{ or } \dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{f} \quad (1)$$

The **phase portrait** of a system of first-order differential equations is a graphical representation that illustrates the behavior of the solutions $\mathbf{x}(t)$ in the state space. Each curve in the phase portrait, called a **trajectory**, corresponds to a solution that starts from a specific initial condition $\mathbf{x}(0)$. The phase portrait is plotted in $\mathbb{R}^n$, where each axis represents one component of the solution vector $\mathbf{x}(t)$. Thus, the solution traces out a curve in $\mathbb{R}^n$ as t varies. The trajectory shows how the solution evolves over time in the state space.

- When $\mathbf{u}$ is in the null space of $\mathbf{B}$ or $\mathbf{u} = \mathbf{0}$, the system of differential equations reduce to

$$\dot{\mathbf{x}} = \mathbf{Ax} \quad (2)$$

If $\mathbf{x}_h$ denotes a solution to system of differential equations (2), then $\alpha\mathbf{x}_h$ is also a solution (proof is quite trivial and left as an exercise). Thus, the system of differential equations of the form (2) constitute a [homogeneous system of differential equations](#).

- **Existence and Uniqueness of Solution to Initial Value Problem:** Let $\mathbf{A}$ and $\mathbf{f}$ be continuous on an interval I . Then, for any $t \in I$, the initial value problem $\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{f}$ and $\mathbf{x}(t_0) = \mathbf{x}_i$ has a unique solution on I .
- **Fundamental set of solutions:** The set $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ constitutes a fundamental set of solution to (2), if they satisfy (2) and are linearly independent.
- The [Wronskian](#) of a set of solutions $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ to the system of differential equations $\dot{\mathbf{x}} = \mathbf{Ax}$ is defined as

$$W = \det \left\{ \underbrace{\begin{bmatrix} \mathbf{x}_1 & \dots & \mathbf{x}_n \end{bmatrix}}_{\mathbf{x}} \right\} \quad (3)$$

- The Wronskian of a set of solutions $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ to $\dot{\mathbf{x}} = \mathbf{Ax}$ satisfies the following properties (see classroom scribble):
 - The Wronskian is either identically 0 or non-zero over the interval $\mathcal{I}$.
 - If $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ constitutes a linearly dependent set, then its Wronskian is identically 0 on $\mathcal{I}$.

- The solution set $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ is linearly independent if the Wronskian is non-zero ($W \neq 0$) at any point in the domain $\mathcal{I}$.

- **Existence of Fundamental Set:** A linear homogeneous system of ODEs $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ always has a fundamental set of solutions on I .

Proof: Consider n initial value problems (IVPs)

$$\begin{aligned}\dot{\mathbf{x}} &= \mathbf{A}\mathbf{x}, & \mathbf{x}(t_0) &= \mathbf{e}_1, \\ \dot{\mathbf{x}} &= \mathbf{A}\mathbf{x}, & \mathbf{x}(t_0) &= \mathbf{e}_2, \\ & \vdots \\ \dot{\mathbf{x}} &= \mathbf{A}\mathbf{x}, & \mathbf{x}(t_0) &= \mathbf{e}_n,\end{aligned}\tag{4}$$

where $\mathbf{e}_i$ is the i -th standard basis vector in $\mathbb{R}^n$, and $t_0 \in I$. Let $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ be the unique solutions to the IVPs (4). Define the matrix of solutions

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1(t) & \mathbf{x}_2(t) & \cdots & \mathbf{x}_n(t) \end{bmatrix}.\tag{5}$$

The Wronskian determinant of these solutions is given by

$$W(\mathbf{X})(t) = \det(\mathbf{X}).\tag{6}$$

For the initial condition t_0 , $\mathbf{X}(t_0) = I$ (the identity matrix) implies

$$W(\mathbf{X})(t_0) = \det(I) = 1 \neq 0.\tag{7}$$

Since $W(\mathbf{X})(t)$ is either always zero or never zero, and it is non-zero at $t = t_0$, we conclude that $W(\mathbf{X})(t) \neq 0$ for all $t \in I$. Therefore, the solutions $\mathbf{x}_1, \dots, \mathbf{x}_n$ are linearly independent. Hence, they form a fundamental set of solutions for the system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ on I .

- **General Solution of Homogeneous Linear System:** Every solution $\phi(t)$ of the initial value problem

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}, \quad \mathbf{x}(t_0) = \mathbf{x}_0\tag{8}$$

can be expressed uniquely as a linear combination of the fundamental set of solutions.

Proof: Let $\{\mathbf{x}_1(t), \dots, \mathbf{x}_n(t)\}$ denote the fundamental set of solutions to the homogeneous system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$, as constructed previously. Then, any linear combination

$$\mathbf{x}(t) = c_1\mathbf{x}_1(t) + c_2\mathbf{x}_2(t) + \cdots + c_n\mathbf{x}_n(t)\tag{9}$$

is also a solution to the system, due to linearity.

To determine whether such a combination can satisfy the initial condition $\mathbf{x}(t_0) = \mathbf{x}_0$, define the matrix $\mathbf{X}(t)$ as

$$\mathbf{X}(t) = \begin{bmatrix} \mathbf{x}_1(t) & \mathbf{x}_2(t) & \cdots & \mathbf{x}_n(t) \end{bmatrix}. \quad (10)$$

Then any solution $\mathbf{x}(t)$ can be written as

$$\mathbf{x}(t) = \mathbf{X}(t)\mathbf{c}, \quad (11)$$

where $\mathbf{c} = \begin{bmatrix} c_1 & c_2 & \cdots & c_n \end{bmatrix}^\top$ is a constant vector.

Evaluating at $t = t_0$, we get

$$\mathbf{x}_0 = \mathbf{X}(t_0)\mathbf{c}. \quad (12)$$

Since $\mathbf{X}(t_0)$ is a full rank matrix and $\mathbf{c}$ can be determined uniquely. Thus the solution to the given system of first-order ODE has the following solution

$$\mathbf{x}(t) = \mathbf{X}(t)\mathbf{x}_0 = \sum_{i=1}^n c_i \mathbf{x}_i(t), \quad (13)$$

which is a linear combination of the fundamental set of solutions. By the uniqueness theorem for systems of first-order ODEs, this is the only solution that satisfies the initial condition. Hence, any solution $\phi(t)$ to the initial value problem must be equal to $\mathbf{x}(t)$, and thus must also be expressible as the same linear combination of the fundamental solutions.